

Assignment /

Assignment #: 19

Topic : Functions

Level 1: Easy

1. Write a function to Calculate the Sum of Odd & Even Numbers
2. Write a function to find the prime numbers between 1 to 100.
3. Get the salary per month and hike from user and the write a C program to calculate new salary with hike.
Hint: The formula is $\text{Salary hike} = \text{oldSalaryPerMonth} + (\text{oldSalaryPerMonth} \times \text{hike}/100)$

Level 2: Medium

1. Write a Python program to find the sums of the running odd numbers and even numbers from a given lowerbound to an upperbound. Also compute their absolute difference. **(Use separate function to find the sum of odd and even numbers from lowerbound to upperbound)**

Sample Input and Output

The sum of odd numbers from 1 to 1000 is: 250000

The sum of even numbers from 1 to 1000 is: 250500

The absolute difference between the two sums is: 500

2. To speed up his composition of generating unpredictable rhythms, Aniruth wants the list of prime numbers available in a range of numbers. Can you help him out? Write a Python program to print all prime numbers in the interval [x,y] (x and y, both inclusive). (Use separate function to print all prime numbers in the given interval)

Sample Input 1:

2

15

Sample Output 1:

2 3 5 7 11 13

Sample Input 2:

8

5

Sample Output 2:

Provide valid input

Sample Input 3:

1

10

Sample Output 3:

2 3 5 7

Level 3: Hard

1. XYZ Technologies is in the process of increment the salary of the employees. This increment is done based on their salary and their performance appraisal rating.
 - a. If the appraisal rating is between 1 and 4, the increment is 10% of the salary.
 - b. If the appraisal rating is between 4.1 and 7, the increment is 25% of the salary.
 - c. If the appraisal rating is between 7.1 and 10, the increment is 30% of the salary.Help them to do this, by writing a program that displays the incremented salary.

(Use separate function to calculate the increment)

Note : If either the salary is 0 or negative (or) if the appraisal rating is not in the range 1 to 10 (inclusive), then the output should be "Invalid Input".

Sample Input 1 :

Enter the salary 80000

Enter the appraisal rating 3

Sample Output 1 :

88000

Sample Input 2 :

Enter the salary 75000

Enter the Performance appraisal rating 4.3

Sample Output 2 :

93750

2. You are developing a modular calculator program in Python, and you want to implement a feature that allows users to perform various mathematical operations dynamically. Your calculator program supports three basic mathematical operations: addition, subtraction, and multiplication.

Your goal is to implement the callback function, which takes a function pointer (operation) representing one of the mathematical operations, along with two integers

(operand1 and operand2). The callback function should dynamically invoke the specified operation on the given operands.

Sample Input:

Add 5 and 8.

Subtract 15 from 20.

Multiply 7 and 9.

Sample Output:

Add: 13

Subtract: 5

Multiply: 63

3. You need to implement a function that finds the maximum value among a variable number of integers. To achieve this flexibility, you decide to use a variadic function. The findMax function takes an integer num representing the number of integers to follow, and a variable number of integers as its arguments. It returns the maximum value among the provided integers. Your goal is to implement the findMax function and demonstrate its usage in the main function.

Sample Input:

maximum value among four integers: 25, 12, 18, and 30.

maximum value among five integers: 8, 15, 22, 17, and 12.

Sample Output:

maximum value among four integers:30

maximum value among five integers:22